Mushfique Shahriar Aurnab

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Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Daegu, Republic of Korea
B.S. student intending to major in Computer Science and Electrical Engineering — *Aug. 2024 – Present*

Maastricht University, Center for European Studies — Maastricht, Netherlands
Summer Programme in European Studies — *June – July 2026*

Research Experience

Soft Augmented Intelligence Lab (SAIL), DGIST — Advisor: Prof. Eojin Rho
Undergraduate Researcher — *Aug. 2026 – Present*

- Surveying foundation-model and world-model papers and assessing how they could be applied to soft wearable robots.
- Designing a simulation study of reinforcement learning for a hip exosuit aimed at shortening 200 m sprint time, using a musculoskeletal human in MyoAssist (MuJoCo). Wrote the plan for reproducing staged human–exoskeleton co-adaptation (SMAT), with matched simple-controller baselines and pass/fail criteria fixed before training.
- Set up the lab's shared Ubuntu GPU server as its temporary administrator: user accounts, key-based SSH over Tailscale, and a setup and user manual for lab members.

Undergraduate Group Research Program (UGRP), DGIST — Advisor: Prof. Kisub Kim
Undergraduate Researcher, Comparative Study of Catastrophic Forgetting in ViTs and CNNs — *Spring 2026 – Present*

- Ran nine sequential CIFAR-10 experiments (three class splits, three seeds) with linear probes at six depths. Probes recovered 79.7% of the old-task examples the final classifier got wrong, and retraining only that classifier restored 74.6% old-task accuracy. None of four targeted replay-selection methods beat random selection.
- Tested whether one adapter reused across 20 CIFAR-100 tasks on a frozen CLIP ViT-B/16 gradually loses the ability to learn new tasks, with the pass/fail gate fixed in advance. LoRA declined by only 3.2–3.6 points, and neither adapter type met the gate.
- Traced the one collapsed run to a single minibatch. From the same checkpoint, 2 of 14 batch orders failed; a bisection placed the damage at update 32, where feature effective rank fell to 18.0 (204.5 in the healthy run), and moving that one batch removed the failure.

Computer Vision Lab, DGIST — Daegu, Republic of Korea
Student Researcher, Video Annotation — *May 2025 – Aug. 2025*

- Labeled video data for video-text alignment and text-based object detection experiments in Prof. Sunghoon Im's lab.

Personal Projects

GPX Motion Analysis | *Python, time-series alignment, Matplotlib* — 2026

- Built a command-line tool that aligns GPX tracks recorded at different times without filling long gaps or connecting separate track segments.
- Added CSV exports, motion metrics, stop-threshold comparisons, JSON summaries, and plots. The tests use reproducible synthetic tracks and run in CI on Python 3.11 and 3.14.

Vox Pop | *Python, MCP, FastEmbed* — 2026

- Built a Python library, command-line tool, and MCP server that searches nine public discussion sites. For each query, a semantic router selects the most relevant communities first.

MapleStory Code Battle AI | *Python, OpenCV, minimax* — 2025

- Reconstructed the game board from screen captures using template matching and color detection. Scored possible moves with summed-area tables and searched three plies ahead using minimax with alpha-beta pruning.

Technical Skills

Languages: Python, C, C++
Machine learning: PyTorch, torchvision, scikit-learn, pandas, NumPy, OpenCV, Matplotlib; linear probing, LoRA and bottleneck adapters
Systems and tools: Linux server administration (Ubuntu, SSH, Tailscale, systemd), Git, GitHub Actions, pytest, LaTeX, MCP servers